

This question paper consists of 8 printed pages, each of which is identified by the Code Number COMP5450M.

A non-programmable calculator may be used.
Answer All Questions.
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School of Computing

January 2019

COMP5450M

KNOWLEDGE REPRESENTATION AND REASONING (MSc)

Time allowed: 2 hours

PLEASE DO NOT REMOVE THIS PAPER FROM THE EXAM ROOM

Answer ALL THREE questions

The marks available for each part of each question are clearly indicated.

Question 1

(a) Translate the following sentence into *Propositional Logic*:

- I go shopping on Mondays and Tuesdays.

[2 marks]

Answer: $(Monday \vee Tuesday) \rightarrow Shopping.$

(b) Translate the following sentences into *First-Order Predicate Logic* (using equality where necessary):

- (i) Some yellow frogs are poisonous.

[2 marks]

Answer: $\exists x[Frog(x) \wedge Yellow(x) \wedge Poisonous(x)]$

- (ii) All Helen's rabbits are white or grey.

[2 marks]

Answer: $\forall x[(Rabbit(x) \wedge Owns(helen, x)) \rightarrow (White(x) \vee Grey(x))]$

- (iii) No dog ate more than one biscuit.

[2 marks]

Answer: $\neg \exists x \exists y \exists z [Dog(x) \wedge Biscuit(y) \wedge Biscuit(z) \wedge \neg(y = z) \wedge Ate(x, y) \wedge Ate(x, z)]$

- (iv) Edward hates everyone except himself.

[2 marks]

Answer: $\forall x[Hates(ed, x) \leftrightarrow \neg(x = ed)]$

(c) $\mathcal{M} = \langle \mathcal{D}, \delta \rangle$ is a model for a first-order language with two unary predicates P and Q and a binary relation predicate R . The domain of $\mathcal{M}$ is the set $\{a, b, c, d, e, f\}$, and the denotation of the predicates is:

- $\delta(P) = \{a, b, c\}$
- $\delta(Q) = \{d, e, f\}$
- $\delta(R) = \{\langle a, f \rangle, \langle b, e \rangle, \langle c, d \rangle, \langle f, f \rangle\}$

Which of the following formulae are satisfied by this model?

[4 marks]

F1. $\forall x[P(x) \vee Q(x)]$

F2. $\exists w[P(w) \wedge Q(w)]$

F3. $\forall x[P(x) \rightarrow \exists y[R(x, y) \wedge Q(y)]]$

F4. $\neg \exists x \exists y[Q(x) \wedge Q(y) \wedge R(x, y)]$

Answer: F1 yes, F2 no, F3 yes, F4 no. 1 mark each.

(d) Use the *Sequent Calculus* to show that the following sequent is valid: **[6 marks]**

$$\forall x[P(x)], \forall x[P(x) \rightarrow Q(x)] \vdash \forall x[Q(x)]$$

You should only use rules from the following rule set, which was presented in the lecture slides, to construct your proof:

$$\frac{\text{Axiom}}{\alpha, \Gamma \vdash \alpha, \Delta}$$

$$\frac{\alpha, \beta, \Gamma \vdash \Delta}{(\alpha \wedge \beta), \Gamma \vdash \Delta} [\wedge\vdash] \quad \frac{\Gamma \vdash \alpha, \Delta \text{ and } \Gamma \vdash \beta, \Delta}{\Gamma \vdash (\alpha \wedge \beta), \Delta} [\wedge\vdash]$$

$$\frac{\alpha, \Gamma \vdash \Delta \text{ and } \beta, \Gamma \vdash \Delta}{(\alpha \vee \beta), \Gamma \vdash \Delta} [\vee\vdash] \quad \frac{\Gamma \vdash \alpha, \beta, \Delta}{\Gamma \vdash (\alpha \vee \beta), \Delta} [\vdash\vee]$$

$$\frac{\Gamma \vdash \alpha, \Delta}{\neg\alpha, \Gamma \vdash \Delta} [\neg\vdash] \quad \frac{\Gamma, \alpha \vdash \Delta}{\Gamma \vdash \neg\alpha, \Delta} [\vdash\neg]$$

$$\frac{\Gamma, \neg\alpha \vee \beta \vdash \Delta}{\Gamma, \alpha \rightarrow \beta \vdash \Delta} [\rightarrow\vdash r.w.] \quad \frac{\Gamma \vdash \neg\alpha \vee \beta, \Delta}{\Gamma \vdash \alpha \rightarrow \beta, \Delta} [\vdash\rightarrow r.w.]$$

$$\frac{\forall x[\Phi(x)], \Phi(k), \Gamma \vdash \Delta}{\forall x[\Phi(x)], \Gamma \vdash \Delta} [\forall\vdash] \quad \frac{\Gamma \vdash \Phi(k), \Delta}{\Gamma \vdash \forall x[\Phi(x)], \Delta} [\vdash\forall]^\dagger$$

† where κ cannot occur anywhere in the lower sequent.

Answer: The sequent is valid as shown by the following proof:

$$\begin{array}{c} \text{AXIOM} \\ \hline \text{Ax}[Px], Pa, \quad \text{Ax}[Px \rightarrow Qx], \quad |- \quad Pa, Qa \\ \hline \text{Ax}[Px], Pa, \quad \text{Ax}[Px \rightarrow Qx], \quad -Pa \quad |- \quad Qa \quad \quad \quad \text{AXIOM} \\ \hline \text{Ax}[Px], Pa, \text{Ax}[Px \rightarrow Qx], Qa \quad |- \quad Qa \\ \hline \text{Ax}[Px], Pa, \text{Ax}[Px \rightarrow Qx], -Pa \vee Qa \quad |- \quad Qa \\ \hline \text{Ax}[Px], Pa, \text{Ax}[Px \rightarrow Qx], Pa \rightarrow Qa \quad |- \quad Qa \\ \hline \text{Ax}[Px], Pa, \text{Ax}[Px \rightarrow Qx] \quad |- \quad Qa \\ \hline \text{Ax}[Px], \text{Ax}[Px \rightarrow Qx] \quad |- \quad Qa \\ \hline \text{Ax}[Px], \text{Ax}[Px \rightarrow Qx] \quad |- \quad \text{Ax}[Qx] \end{array}$$

There is basically a mark for each correct rule application. Some credit may be given for almost correct rule applications.

[Question 1 total: 20 marks]

Answers

Question 2

- (a) (i) Give the set of *clausal* formulae (i.e. formulae in *disjunctive normal form*)¹ corresponding to the following propositional formulae: [4 marks]

$$\neg\neg A \vee S, (\neg S \wedge T), (A \vee B) \rightarrow Q, (Q \wedge T) \rightarrow (R \wedge S)$$

Answer:

$$\{ \{A, S\}, \{\neg S\}, \{T\}, \{\neg A, Q\}, \{\neg B, Q\}, \{\neg Q, \neg T, R\}, \{\neg Q, \neg T, S\} \}$$

- (ii) Give a proof that these formulae are inconsistent using *binary propositional resolution*. [4 marks]

1.	$\{A, S\}$	8.	$\{A\}$	1&2
2.	$\{\neg S\}$	9.	$\{Q\}$	4&8
3.	$\{T\}$	10.	$\{\neg T, S\}$	7&9
Answer: 4.	$\{\neg A, Q\}$	11.	$\{\neg T\}$	2&10
5.	$\{\neg B, Q\}$	12.	$\emptyset$	3&11
6.	$\{\neg Q, \neg T, R\}$			
7.	$\{\neg Q, \neg T, S\}$			

- (b) Translate the following sentence into *Propositional Tense Logic*: [2 marks]

If I win the lottery I will be rich forever after that.

Answer: $G(Win \rightarrow GRich)$ (or $G(PWin \rightarrow Rich)$, though not quite right)

- (c) A Situation Calculus theory makes use of fluents of the forms:

$robot_has(item)$ $on_floor(item, room)$ $locked(door)$
 $robot_location(room)$ $connects(door, room_1, room_2)$

The theory includes constants referring to items, one of which is key.

The theory also describes the behaviour of a robot in terms of the following actions:

pick_up(object) **unlock(door)** **move_to(room)**

An initial situation, s_0 , is described as follows:

Holds($connects(door1, hall, lounge)$, s_0) Holds($connects(door2, hall, study)$, s_0)
 $\neg$ Holds($locked(door1)$, s_0) Holds($locked(door2)$, s_0)
Holds($on_floor(key, lounge)$, s_0) Holds($robot_location(hall)$, s_0)

- (i) Assuming that the initial situation is s_0 , give a sequence of actions that will result in the goal $robot_location(study)$ being satisfied. [2 marks]

¹This was an error in the exam script. A set of clausal formulae is more like *conjunctive normal form*, except that we give it as a set of *clauses* rather than a conjunction (the conjunction is implicit rather than explicitly written with a ' $\wedge$ ' symbol). Each clause is either a *literal* or a disjunction of literals. A literal is either an atomic proposition or a negated atomic proposition.

- (ii) For each of the actions `pick_up` and `move_to` specify a *precondition* axiom stating the conditions under which the action is possible. **[4 marks]**
- (iii) Give an *effect* axiom specifying the results of carrying out the action `unlock`. **[2 marks]**
- (iv) Write down a *frame* axiom stating that the `move_to` action does not affect the *locked* fluent. **[2 marks]**

Answer:

- (i) `move(lounge), pickup(key), move(hall), unlock(door2), move(study)`
- (ii)
$$\begin{aligned} \text{-- Poss(pickup}(x),s) &\leftarrow \exists r [\text{Holds(robot_location}(r),s) \ \& \ \text{Holds(Onfloor}(x,r),s)] \\ \text{-- Poss(move}(x),s) &\leftarrow \exists r \ d [\text{Holds(connects}(d,r,x),s) \ \& \ \text{Holds(robot_location}(r),s) \ \& \ \text{-- Holds(locked}(d),s)] \\ &\quad . \end{aligned}$$
- (It is actually also ok to leave out the existential quantifier in these axioms, since the usual implicit universal quantification converts to existential when on right of $\leftarrow$.)
- (iii)
$$\text{-- -- Holds(locked}(x), \text{result(unlock}(x),s)) \leftarrow \text{Poss(unlock}(x),s)$$
- (v)
$$\text{Holds(locked}(x), \text{result(move}(x),s)) \leftrightarrow \text{Holds(locked}(x),s)$$

[Question 2 total: 20 marks]

Question 3

(a) For each of the following *Prolog* queries, give the value of the variable X after the query has been executed:

(i) ?- $X = 7/2$. [1 mark]

(ii) ?- $[1, [2, 3], 4] = [_ | [X | _]]$ [1 mark]

(iii) ?- $A = [1, 2, 3, 4, 5], \text{setof}(I, (\text{member}(I, A), I > 2), X)$. [1 mark]

(iv) ?- $\text{append}([X], [2, 3], [1, 2, 3])$. [1 mark]

Answer:

(i) $7/2$

(ii) $[2, 3]$

(iii) $[3, 4, 5]$

(iv) 1

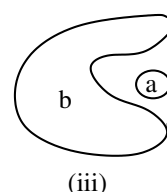
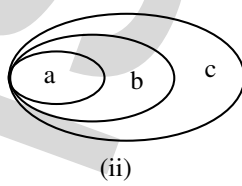
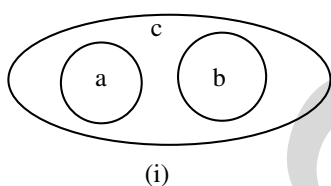
(b) Consider the following formulae involving topological relations of the *Region Connection Calculus* (RCC) and the *convex hull* function, conv . The constants (a , b and c) refer to particular spatial regions. In each case, draw a configuration of the regions referred to by these constants that satisfies the formula, labelling your diagram to indicate which region is which:

(i) $\text{DC}(a, b) \wedge \text{NTPP}(\text{sum}(a, b), c)$ [2 marks]

(ii) $\text{TPP}(a, b) \wedge \text{TPP}(b, c) \wedge \text{TPP}(a, c)$ [2 marks]

(iii) $\text{DC}(a, b) \wedge \text{TPP}(a, \text{conv}(b))$ [2 marks]

Answer: Possible diagrams are as follows:



(c) A *liger* is an animal whose parents are a male lion and a female tiger. Use *Description Logic* to give a definition of the concept **Liger** in terms of the concepts **Lion**, **Tiger**, **Male**, **Female** and the relation **hasParent**. [4 marks]

Answer:

$$\text{Liger} \equiv \exists \text{hasParent}.(\text{Male} \sqcap \text{Lion}) \sqcap \exists \text{hasParent}.(\text{Female} \sqcap \text{Tiger})$$

- (d) Write a *Default Logic* rule that formally represents the reasoning principle expressed in the following statement: **[2 marks]**

“British people typically drink tea, except for children and those who drink coffee.”

Answer: $British(x) : Drinks(x, tea), \neg Child(x), \neg Drinks(x, coffee) / Drink(x, tea)$

- (e) This question concerns a *Fuzzy Logic* in which the following definitions of *linguistic modifiers* are specified:

$$\text{quite}(\phi) = \phi^{1/2} \quad \text{very}(\phi) = \phi^2$$

The logic is used to describe Leo the lion, who possesses certain characteristics to the following degrees:

$$\text{Large}(\text{leo}) = 0.5 \quad \text{Fierce}(\text{leo}) = 0.09 \quad \text{Clever}(\text{leo}) = 0.4$$

Translate the following sentences into fuzzy logic and also give the fuzzy truth value of each proposition (under the standard fuzzy interpretation of the Boolean connectives):

- (i) Leo is not very clever. **[2 marks]**
(ii) Leo is very very large and quite fierce. **[2 marks]**

Answer:

A) $\neg \text{very}(\text{Clever}(\text{leo}))$ (1 mark)

Truth value $1 - (0.4)^2 = 1 - 0.16 = 0.84$ (1 mark)

B) $\text{very}(\text{very}(\text{Large}(\text{leo}))) \wedge \text{quite}(\text{Fierce}(\text{leo}))$ (1 mark)

Truth value = $\text{Min}(((0.5)^2)^2, (0.09)^{\frac{1}{2}}) = \text{Min}(0.0625, 0.3) = 0.0625$ (1 mark)

[Question 3 total: 20 marks]

[Grand total: 60 marks]